

WEB INFRASTRUCTURE DOCUMENTATION

1. Establishing the Environment

The practical was completed on one Linux machine. The administrator account was used for system-level checks. The machine uses Parrot Security OS 7.3 with hostname parrot. The loopback address 127.0.0.1 was used for local client/server testing.

- Operating system: Parrot Security OS 7.3
- Kernel: Linux 7.0.9+parrot7-amd64
- Hostname: parrot
- Loopback: 127.0.0.1/8
- Wireless interface: 192.168.0.214/16

```
(leo_01@parrot) ~$ whoami
leo_01
(leo_01@parrot) ~$ hostname
parrot
(leo_01@parrot) ~$ hostnamectl
Static hostname: parrot
Icon name: computer-laptop
Chassis: laptop
Machine ID: 1d14e86ccf5244248fc003c291601045
Boot ID: eadb03bf3bd44893b3b33e5ec9e3bd41
Operating System: Parrot Security 7.3 (echo)
Kernel: Linux 7.0.9+parrot7-amd64
Architecture: x86_64
Hardware Vendor: Dell Inc.
Hardware Model: Dell 15 DC15250
Firmware Version: 1.11.0
Firmware Date: Wed 2026-05-27
Firmware Age: 3month 1w 3d
(leo_01@parrot) ~$ hostname -I
192.168.0.214
(leo_01@parrot) ~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 48:45:66:6a:8c:33 brd ff:ff:ff:ff:ff:ff
    altname wlx4845666a8c33
    inet 192.168.0.214/16 brd 192.168.255.255 scope global dynamic noprefixroute wlp1s0
        valid_lft 35715sec preferred_lft 35715sec
    inet6 fe80::1a95:b32d:9554:a3d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(leo_01@parrot) ~$ cat /etc/os-release
PRETTY_NAME="Parrot Security 7.3 (echo)"
NAME="Parrot Security"
VERSION_ID="7.3"
VERSION="7.3 (echo)"
VERSION_CODENAME=echo
DEBIAN_VERSION_FULL=13.5
ID=parrot
ID_LIKE=debian
HOME_URL="https://www.parrotsec.org/"
SUPPORT_URL="https://www.parrotsec.org/docs"
BUG_REPORT_URL="https://github.com/parrotsec/"
```

3. Before doing anything: establish your environment

Open a terminal using your normal Parrot administrator account.

Use:

```
0: Bash
whoami
hostname
hostnamectl
hostname -I
ip addr
```

Also check the OS:

```
0: Bash
```

You should record:

Distribution ID	Hostname	IP address	Client username	Loopback

The assignment specifically asks you to document the distribution, IP, and why localhost/127.0.0.1 is being used.

Why 127.0.0.1?

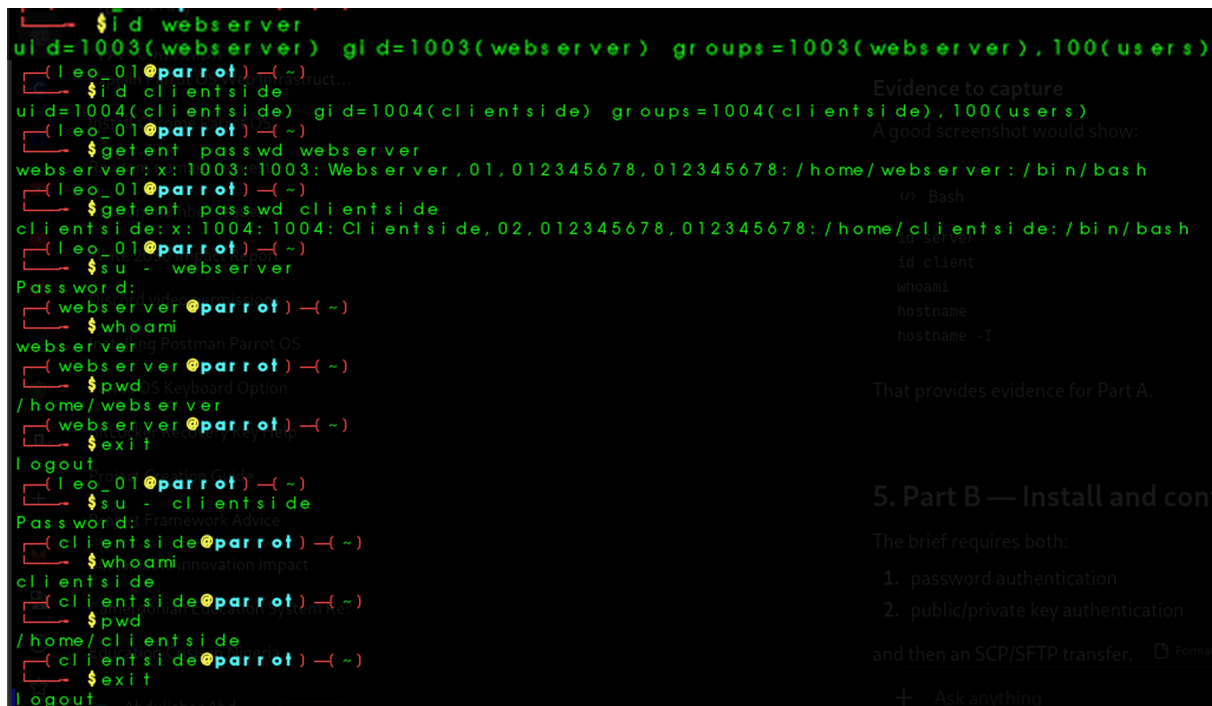
127.0.0.1 is the loopback address.

It means:

2. Verifying the Two Accounts

The two required accounts were checked and their home directories were verified. The evidence shows separate accounts for webserver and clientside, with the administrator account used for system-level work.

- webserver: UID 1003, home /home/webserver
- clientside: UID 1004, home /home/clientside



```
$ id webserver
uid=1003(webserver) gid=1003(webserver) groups=1003(webserver),100(users)
leo_01@parrot:~$ id clientside
uid=1004(clientside) gid=1004(clientside) groups=1004(clientside),100(users)
leo_01@parrot:~$ getent passwd webserver
webserver:x:1003:1003:Webserver,01,012345678,012345678:/home/webserver:/bin/bash
leo_01@parrot:~$ getent passwd clientside
clientside:x:1004:1004:Clientside,02,012345678,012345678:/home/clientside:/bin/bash
leo_01@parrot:~$ su - webserver
Password:
webserver@parrot:~$ whoami
webserver
webserver@parrot:~$ pwd
/home/webserver
webserver@parrot:~$ exit
logout
leo_01@parrot:~$ su - clientside
Password:
clientside@parrot:~$ whoami
clientside
clientside@parrot:~$ pwd
/home/clientside
clientside@parrot:~$ exit
logout
```

Evidence to capture
A good screenshot would show:

- id server
- id client
- whoami
- hostname
- hostname -I

That provides evidence for Part A.

5. Part B — Install and configure

The brief requires both:

1. password authentication
2. public/private key authentication

and then an SCP/SFTP transfer.

+ Ask anything

3. Password-Based SSH Login

Password authentication was tested first to establish a baseline. The SSH session connected to the local webserver account on 127.0.0.1. The SSH service was confirmed to be running before the connection was tested. The SSH server configuration is stored in `/etc/ssh/sshd_config`. The configuration allows the SSH server to listen for incoming connections and controls authentication methods. Password authentication was used first to confirm that the `clientside` account could connect to the `webserver` account on the same machine through `127.0.0.1`. This demonstrates a client-server SSH connection even though both accounts are hosted on the same Linux machine.

[illegible]

5. Deploying the Public Key

```
(clientside@parrot) ~$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/clientside/.ssh/id_ed25519):
Enter passphrase for '/home/clientside/.ssh/id_ed25519' (empty for no passphrase):
Enter same passphrase again:
Passphrases do not match. Try again.
Enter passphrase for '/home/clientside/.ssh/id_ed25519' (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/clientside/.ssh/id_ed25519
Your public key has been saved in /home/clientside/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:txAdnigInlCgJqk5JV6l4M8xd48petVw0mf7R6iknnQ clientside@parrot
The key's randomart image is:
+--[ED25519 256]--+
|      .            |
|    o + o          |
|   + + + o         |
|  = o o c S x      |
|   + . + +         |
|..o + o + E o      |
|+.o + o .+Eo       |
|-----+ (SHA256) -----
[clientside@parrot] (~)
$ ls -la ~/.ssh
total 16
drwx----- 1 clientside clientside 100 Sep  5 19:57 .
drwx----- 1 clientside clientside 304 Sep  5 19:47 ..
-rw----- 1 clientside clientside 464 Sep  5 19:57 id_ed25519
-rw-r--r-- 1 clientside clientside  99 Sep  5 19:57 id_ed25519.pub
-rw-r--r-- 1 clientside clientside 978 Sep  5 19:52 known_hosts
-rw-r--r-- 1 clientside clientside 142 Sep  5 19:47 known_hosts.old
[clientside@parrot] (~)
$ ssh-copy-id webserver@127.0.0.1
/bin/ssh-copy-id: INFO: Source of key(s) to be installed: '/home/clientside/.ssh/id_ed25519.pub'
/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install
webserver@127.0.0.1's password:
Number of key(s) added: 1
Now try logging into the machine, with: 'ssh 'webserver@127.0.0.1'
and check to make sure that only the key(s) you wanted were added.
(webserver@parrot) ~$ ssh
$ ls -ld ~/.ssh
drwx----- 1 webserver webserver 30 Sep  5 20:01 /home/webserver/.ssh
(webserver@parrot) ~$ ls -ld ~/.ssh/authorized_keys
-rw-r--r-- 1 webserver webserver 99 Sep  5 20:01 /home/webserver/.ssh/authorized_keys
(webserver@parrot) ~$ chmod 700 ~/.ssh
(webserver@parrot) ~$ chmod 600 ~/.ssh/authorized_keys
(webserver@parrot) ~$ ls -ld ~/.ssh
drwx----- 1 webserver webserver 30 Sep  5 20:01 /home/webserver/.ssh
(webserver@parrot) ~$ ls -ld ~/.ssh/authorized_keys
-rw-r--r-- 1 webserver webserver 99 Sep  5 20:01 /home/webserver/.ssh/authorized_keys
(webserver@parrot) ~$ exit
logout
Connection to 127.0.0.1 closed.
```

The public key was copied to the webserver account and the SSH directory permissions were checked. The evidence shows the .ssh directory and authorized_keys file with restricted permissions. Correct permissions on the .ssh directory and authorized_keys are important because SSH may reject key authentication when these files are accessible by unauthorized users.

6. Confirming Key-Based Login

Verbose SSH output was used to verify key authentication. The client offered the Ed25519 public key, the server accepted it, and authentication completed using `publickey`.


```
debug1: Offering public key: /home/client/de/.ssh/id_ed25519 ED25519 SHA256:txAdnI glnlCgJqk6Jv6l4M8xd48petVw0mf7R6i knnQ
debug1: Server accepts key: /home/client/de/.ssh/id_ed25519 ED25519 SHA256:txAdnI glnlCgJqk6Jv6l4M8xd48petVw0mf7R6i knnQ
Enter passphrase for key '/home/client/de/.ssh/id_ed25519':
Authenticated to 127.0.0.1 ((127.0.0.1):22) using "publickey".
debug1: channel 0: new session (client-session) (inactive timeout: 0)
debug1: Requesting no-more-sessions@openssh.com
debug1: Entering interactive session.
debug1: pledge: filesystem
debug1: client_input_global_request: rtype hostkeys-00@openssh.com want_reply 0
debug1: client_input_hostkeys: searching /home/client/de/.ssh/known_hosts for 127.0.0.1 / (none)
debug1: client_input_hostkeys: searching /home/client/de/.ssh/known_hosts for 127.0.0.1 / (none)
debug1: client_input_hostkeys: hostkeys: file /home/client/de/.ssh/known_hosts2 does not exist
debug1: client_input_hostkeys: no new or deprecated keys from server
debug1: Remote: /home/webserver/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Remote: /home/webserver/.ssh/authorized_keys:1: key options: agent-forwarding port-forwarding pty user-rc x11-forwarding
debug1: Sending environment.
debug1: channel 0: setting env LC_ADDRESS = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_NAME = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_MONETARY = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_PAPER = 'en_GB.UTF-8'
debug1: channel 0: setting env LANG = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_IDENTIFICATION = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_TELEPHONE = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_MEASUREMENT = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_TIME = 'en_GB.UTF-8'
debug1: channel 0: setting env LC_NUMERIC = 'en_GB.UTF-8'
debug1: pledge: park
Linux parrot 7.0.9+parrot7-amd64 #1 SMP PREEMPT_DYNAMIC Parrot 7.0.9-1parrot1 (2026-05-28) x86_64
Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.
Last login: Sat Sep 5 20:01:28 2026 from 127.0.0.1

13. SFTP demonstration
Inside SFTP:
sftp> pwd
/sftp-server@127.0.0.1
sftp> ls
transfer-test.txt
sftp> get transfer-test.txt
100% 30 149.3KB/s 00:00
sftp> bye
Interactive file transfer protocol over SSH
```

7. SCP File Transfer

A test file was transferred from the clientside account to the webserver account using SCP. The receiving side was then checked to confirm the transfer.

```
(client/de)parrot~$ echo "SSH transfer test from client" > ~/transfer-test.txt
(client/de)parrot~$ scp ~/transfer-test.txt webserver@127.0.0.1:/home/webserver/
Enter passphrase for key '/home/client/de/.ssh/id_ed25519':
(client/de)parrot~$ sftp webserver@127.0.0.1
Enter passphrase for key '/home/client/de/.ssh/id_ed25519':
Linux parrot 7.0.9+parrot7-amd64 #1 SMP PREEMPT_DYNAMIC Parrot 7.0.9-1parrot1 (2026-05-28) x86_64

13. SFTP demonstration
Inside SFTP:
sftp> pwd
/sftp-server@127.0.0.1
sftp> ls
transfer-test.txt
sftp> get transfer-test.txt
100% 30 149.3KB/s 00:00
sftp> bye
Interactive file transfer protocol over SSH

The program included with the Parrot GNU/Linux are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*copyright.
Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 5 20:25:31 2026 from 127.0.0.1
(webserver)parrot~$ ls -l /home/webserver/transfer-test.txt
-rw-r--r-- 1 webserver webserver 30 Sep 5 20:28 /home/webserver/transfer-test.txt
(webserver)parrot~$ cat /home/webserver/transfer-test.txt
SSH transfer test from client
(webserver)parrot~$ sftp webserver@127.0.0.1
Enter passphrase for key '/home/client/de/.ssh/id_ed25519':
Connected to 127.0.0.1.
sftp> pwd
Remote working directory: /home/webserver
sftp> ls
Desktop  Documents  Downloads  Music  Pictures  Templates  Videos  transfer-test.txt
sftp> put ~/transfer-test.txt
stat ~/transfer-test.txt: No such file or directory
sftp> put ~/transfer-test.txt
stat ~/transfer-test.txt: No such file or directory
sftp> put /home/webserver/transfer-test.txt
put /home/webserver/transfer-test.txt: Permission denied
sftp> sudo put /home/webserver/transfer-test.txt
Invalid command.
sftp> get transfer-test.txt
get /home/webserver/transfer-test.txt to transfer-test.txt
transfer-test.txt
sftp> put ~/transfer-test.txt
stat ~/transfer-test.txt: No such file or directory
sftp> ls
Desktop  Documents  Downloads  Music  Pictures  Templates  Videos  transfer-test.txt
sftp> pwd
Remote working directory: /home/webserver
sftp> put /home/
put /home/: No such file or directory
sftp> bye
```

8. SFTP File Transfer

An SFTP session was used to check the remote and local working directories and to upload the test file. The evidence also shows the permission boundary when an upload was attempted outside the permitted location.

9. NGINX Installation and Configuration Review

The NGINX configuration directories were reviewed. The evidence includes `/etc/nginx`, `sites-available`, `sites-enabled`, the web root and the NGINX log directory. The NGINX configuration uses several important directives. `listen` defines the port on which the server accepts HTTP requests, normally port 80. `server_name` identifies the hostname handled by the server block. `root` specifies the directory containing the website files, while `index` defines the default file served when a directory is requested. The site configuration is stored in `sites-available` and enabled through `sites-enabled`. The `nginx -t` command was used to verify that the configuration syntax was valid before the service was used

```

(léo_01@parrot) (~)
$ ls -la /etc/nginx
total 48
drwxr-xr-x 1 root root 352 Sep 5 20:49 .
drwxr-xr-x 1 root root 5460 Sep 5 19:32 ..
drwxr-xr-x 1 root root 0 Jun 5 16:22 conf.d
-rw-r--r-- 1 root root 1125 Jun 5 16:22 fastcgi.conf
-rw-r--r-- 1 root root 2460 Jun 5 16:22 fastcgi_params
-rw-r--r-- 1 root root 2837 Jun 5 16:22 koi-utf
-rw-r--r-- 1 root root 2223 Jun 5 16:22 koi-win
-rw-r--r-- 1 root root 5527 Jun 5 16:22 mime.types
drwxr-xr-x 1 root root 0 Jun 5 16:22 modules-available
drwxr-xr-x 1 root root 0 Jun 5 16:22 modules-enabled
-rw-r--r-- 1 root root 1545 Jun 5 16:22 nginx.conf
-rw-r--r-- 1 root root 1545 Jun 5 16:22 proxy_params
-rw-r--r-- 1 root root 2038 Jun 5 16:22 scgi_params
drwxr-xr-x 1 root root 14 Sep 5 20:49 sites-available
drwxr-xr-x 1 root root 14 Jun 25 22:45 sites-enabled
drwxr-xr-x 1 root root 58 Sep 5 20:49 snippets
-rw-r--r-- 1 root root 2067 Jun 5 16:22 uwsgi_params
-rw-r--r-- 1 root root 3071 Jun 5 16:22 win-utf

(léo_01@parrot) (~)
$ ls -la /etc/nginx/sites-available
total 4
drwxr-xr-x 1 root root 14 Sep 5 20:49 .
drwxr-xr-x 1 root root 352 Sep 5 20:49 ..
-rw-r--r-- 1 root root 2412 Jun 5 16:22 default

(léo_01@parrot) (~)
$ ls -la /etc/nginx/sites-enabled/
total 4
drwxr-xr-x 1 root root 14 Jun 25 22:45 .
drwxr-xr-x 1 root root 352 Sep 5 20:49 ..
lrwxrwxrwx 1 root root 34 Jun 25 22:45 default -> /etc/nginx/sites-available/default

(léo_01@parrot) (~)
$ ls -la /var/www
total 0
drwxr-xr-x 1 root root 8 Jun 25 22:45 .
drwxr-xr-x 1 root root 120 Aug 19 15:42 ..
drwxr-xr-x 1 root root 104 Jun 25 22:45 html

(léo_01@parrot) (~)
$ ls -la /var/log/nginx
total 8
drwxr-xr-x 1 root adm 38 Jun 25 22:45 .
drwxr-xr-x 1 root root 868 Sep 5 20:35 ..
-rw-r--r-- 1 www-data adm 88 Sep 5 20:52 access.log
-rw-r--r-- 1 www-data adm 0 Jun 25 22:45 error.log

```

10. NGINX Site Configuration and Website

The NGINX site was enabled and the configuration test completed successfully. The website files and the custom page were checked. The screenshots also show the HTML used for the custom Web Infrastructure Assessment page. The custom website was served from the configured web root and successfully accessed through `localhost`. A custom error page was also configured so that requests for unavailable resources returned the designed error page instead of relying only on the default NGINX page. This confirms that NGINX was serving our own website configuration correctly.

```
(clientside@parrot) ~$ ssh webserver@127.0.0.1
Enter passphrase for key '/home/clientside/.ssh/id_ed25519':
Linux parrot 7.0.9+parrot7-amd64 #1 SMP PREEMPT_DYNAMIC Parrot 7.0.9-1parrot1 (2026-05-28) x86_64

[+] Welcome to the Parrot GNU/Linux distribution!
[+] Project Framework Advice
[+] Sustainable innovation impact

The programs included with the Parrot GNU/Linux are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 5 19:52:32 2026 from 127.0.0.1
(leo_01@parrot) ~$ whoami
leo_01
(leo_01@parrot) ~$ hostname
parrot
(leo_01@parrot) ~$ hostnamectl
Static hostname: parrot
Icon name: computer-laptop
Chassis: laptop
Machine ID: 1d14e86ccf5244248fc003c291601045
Boot ID: eacb03bf3bd44893b3b33e5ec9e3bd41
Operating System: Parrot Security 7.3 (echo)
Kernel: Linux 7.0.9+parrot7-amd64
Architecture: x86-64
Hardware Vendor: Dell Inc.
Hardware Model: Dell 15 DC15250
Firmware Version: 1.11.0
Firmware Date: Wed 2026-05-27
Firmware Age: 3month 1w 3d
(leo_01@parrot) ~$ hostname -I
192.168.0.214
(leo_01@parrot) ~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 48:45:e6:6a:8e:33 brd ff:ff:ff:ff:ff:ff
    altname wlx4845e66a8e33
    inet 192.168.0.214/16 brd 192.168.255.255 scope global dynamic noprefixroute wlp1s0
        valid_lft 35715sec preferred_lft 35715sec
    inet6 fe80::1a95:b32d:9554:a3d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(leo_01@parrot) ~$ cat /etc/os-release
PRETTY_NAME="Parrot Security 7.3 (echo)"
NAME="Parrot Security"
VERSION_ID="7.3"
VERSION="7.3 (echo)"
VERSION_CODENAME=echo
DEBIAN_VERSION_FULL=13.5
ID=parrot
ID_LIKE=debian
HOME_URL="https://www.parrotsec.org/"
SUPPORT_URL="https://www.parrotsec.org/docs"
BUG_REPORT_URL="https://github.com/parrotsec/"

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individual files in /usr/share/doc/*/copyright.

Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Sep 5 19:52:32 2026 from 127.0.0.1
(leo_01@parrot) ~$ whoami
leo_01
(leo_01@parrot) ~$ hostname
parrot
(leo_01@parrot) ~$ hostnamectl
Static hostname: parrot
Icon name: computer-laptop
Chassis: laptop
Machine ID: 1d14e86ccf5244248fc003c291601045
Boot ID: eacb03bf3bd44893b3b33e5ec9e3bd41
Operating System: Parrot Security 7.3 (echo)
Kernel: Linux 7.0.9+parrot7-amd64
Architecture: x86-64
Hardware Vendor: Dell Inc.
Hardware Model: Dell 15 DC15250
Firmware Version: 1.11.0
Firmware Date: Wed 2026-05-27
Firmware Age: 3month 1w 3d
(leo_01@parrot) ~$ hostname -I
192.168.0.214
(leo_01@parrot) ~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: wlp1s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 48:45:e6:6a:8e:33 brd ff:ff:ff:ff:ff:ff
    altname wlx4845e66a8e33
    inet 192.168.0.214/16 brd 192.168.255.255 scope global dynamic noprefixroute wlp1s0
        valid_lft 35715sec preferred_lft 35715sec
    inet6 fe80::1a95:b32d:9554:a3d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(leo_01@parrot) ~$ cat /etc/os-release
PRETTY_NAME="Parrot Security 7.3 (echo)"
NAME="Parrot Security"
VERSION_ID="7.3"
VERSION="7.3 (echo)"
VERSION_CODENAME=echo
DEBIAN_VERSION_FULL=13.5
ID=parrot
ID_LIKE=debian
HOME_URL="https://www.parrotsec.org/"
SUPPORT_URL="https://www.parrotsec.org/docs"
BUG_REPORT_URL="https://github.com/parrotsec/"
```

9. Deploy the public key

The easiest method is:

❖ Bash

ssh-copy-id server@127.0.0.1

Enter the (server) user's password when asked.

Then verify:

3. Before doing anything: establish your environment

Open a terminal using your normal Parrot administrator account.

Use:

❖ Bash

whoami

hostname

hostnamectl

hostname -I

ip addr

Also check the OS:

❖ Bash

You should record:

Host name: _____

IP address: _____

Default username: _____

Loopback: 127.0.0.1

The assignment specifically asks you to document the distribution, IP

and why localhost/127.0.0.1 is being used. [Formative 2 - Web Infrastructure](#)

Why 127.0.0.1?

127.0.0.1 is the loopback address.

It means:

```
/www/webinfrastructure
01:
www/webinfrastructure/index.htm
www/webinfrastructure/custom_40
data: www-data /var/www/webin
www/webinfrastructure -type d -
01:
www/webinfrastructure -type f -
webinfrastructure
structure
www-data 50 Sep 9 18:39 /var/
www-data 235 Sep 9 18:39 cust
www-data 291 Sep 9 18:39 inde
```

```

leo_01@parrot: ~$ ls -l /etc/nginx/
total 44
drwxr-xr-x 1 root root 0 Jun 5 16:22 conf.d
-rw-r--r-- 1 root root 1125 Jun 5 16:22 fastcgi.conf
-rw-r--r-- 1 root root 2460 Jun 5 16:22 fastcgi_params
-rw-r--r-- 1 root root 2837 Jun 5 16:22 koi-utf
-rw-r--r-- 1 root root 2223 Jun 5 16:22 koi-win
-rw-r--r-- 1 root root 5527 Jun 5 16:22 mime.types
drwxr-xr-x 1 root root 0 Jun 5 16:22 modules-available
drwxr-xr-x 1 root root 0 Jun 5 16:22 modules-enabled
-rw-r--r-- 1 root root 1545 Jun 5 16:22 nginx.conf
-rw-r--r-- 1 root root 1545 Jun 5 16:22 proxy_params
-rw-r--r-- 1 root root 2038 Jun 5 16:22 scgi_params
drwxr-xr-x 1 root root 90 Sep 5 22:22 sites-available
drwxr-xr-x 1 root root 20 Sep 5 21:52 sites-enabled
drwxr-xr-x 1 root root 58 Sep 5 20:49 snippets
-rw-r--r-- 1 root root 2067 Jun 5 16:22 uwsgi_params
-rw-r--r-- 1 root root 3071 Jun 5 16:22 win-utf

leo_01@parrot: ~$ ls -l /etc/nginx/sites-available/
total 12
-rw-r--r-- 1 root root 2412 Jun 5 16:22 default
-rw-r--r-- 1 root root 1320 Sep 5 22:26 webinfrastructure
-rw-r--r-- 1 root root 1320 Sep 5 22:21 webinfrastructure.bak

leo_01@parrot: ~$ ls -l /etc/nginx/sites-enabled/
total 4
lrwxrwxrwx 1 root root 44 Sep 5 21:52 group-site -> /etc/nginx/sites-available/webinfrastructure

leo_01@parrot: ~$ ls -l /etc/nginx/sites-enabled/group-site
lrwxrwxrwx 1 root root 44 Sep 5 21:52 /etc/nginx/sites-enabled/group-site -> /etc/nginx/sites-available/webinfrastructure

leo_01@parrot: ~$ less /etc/nginx/sites-enabled/group-site
leo_01@parrot: ~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful

```

```

leo_01@parrot) ~)
$ su - webserver
Password:
leo_01@parrot) ~)
$ curl http://127.0.0.1
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>

  <!-- Makes the website responsive on phones, tablets and computers -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>Web Infrastructure Assessment</title>

  <!-- Link the external CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>

  <!-- Main header -->
  <header class='hero'>
    <div class='container'>

      <p class='server-status'>• NGINX SERVER ONLINE</p>

      <h1>Web Infrastructure Assessment</h1>

      <p class='subtitle'>
        Building and Securing a Mini Production-Ready Web Infrastru
      </p>
    </div>
  </header>

  <!-- Main website content -->
  <main class='container'>

    <!-- About the project -->
    <section class='card'>
      <h2>About This Project</h2>

      <p>
        This website has been created as part of our Web Infrastruc
        assessment. It is hosted locally using the NGINX web server
        running on Parrot OS Linux.
      </p>

      <p>
        The project demonstrates the configuration, testing and
        troubleshooting of essential web infrastructure services,
        including SSH, NGINX and HTTP.
      </p>
    </section>

```

```

HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 15:56:10 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: keep-alive
ETag: '6a9c4efa-106d'
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>

  <!-- Makes the website responsive on phones, tablets and computers -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>Web Infrastructure Assessment</title>

  <!-- Link the external CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
  <!-- Main header -->
  <header class='hero'>
    <div class='container'>

      <p class='server-status'>● NGINX SERVER ONLINE</p>

      <h1>Web Infrastructure Assessment</h1>

      <p class='subtitle'>
        Building and Securing a Mini Production-Ready Web Infrastru
      </p>

    </div>
  </header>

  <!-- Main website content -->
  <main class='container'>

    <!-- About the project -->
    <section class='card'>
      <h2>About This Project</h2>

      <p>
        This website has been created as part of our Web Infrastruc
        assessment. It is hosted locally using the NGINX web server
        running on Parrot OS Linux.
      </p>

```

11. HTTP Methods, Status Codes and Response Headers

The HTTP tests demonstrate different request methods and status codes against the local NGINX service. The evidence shows GET/POST/PUT/DELETE testing, a 404 response for a missing resource, and response headers. The terminal output also records four different status codes: 200, 201, 204 and 404.

GET was used to retrieve a resource, while POST was used to submit data. PUT was used to update or replace a resource, and DELETE was used to request its removal. The tests produced several HTTP status codes. 200 OK indicates that a request was successful, 201 Created indicates successful creation

of a resource, **204 No Content** indicates that the request succeeded without returning a response body, and **404 Not Found** indicates that the requested resource could not be found. The response headers provide additional information about the HTTP response and server.

```
(clientside@parrot) (~)
$ curl -i -X POST -d 'test=1' http://localhost/method-test
HTTP/1.1 201 Created
Server: nginx
Date: Wed, 09 Sep 2026 15:56:33 GMT
Content-Type: text/plain
Content-Length: 20
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

POST method handled
(clientside@parrot) (~)
$ curl -i -X PUT -d 'test=2' http://localhost/method-test
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 15:56:46 GMT
Content-Type: text/plain
Content-Length: 19
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

PUT method handled
(clientside@parrot) (~)
$ curl -i -X DELETE http://localhost/method-test
HTTP/1.1 204 No Content
Server: nginx
Date: Wed, 09 Sep 2026 15:56:55 GMT
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

POST method handled
(clientside@parrot) (~)
$ curl -i http://localhost/does-not-exist
HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 15:57:25 GMT
Content-Type: text/html
Content-Length: 662
Connection: keep-alive
ETag: "6a9c5478-296"
X-Module: Web-Infrastructure
X-Group: Group-07

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>404 - Page Not Found</title>

  <!-- Connect to the existing CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>

<body>

  <main class='error-page'>
```



```
(clientside@parrot) ~)
$ curl -s -o /dev/null -w '%(http_code)\n' http://localhost/
curl -s -o /dev/null -w '%(http_code)\n' -X POST http://localhost/method-test
curl -s -o /dev/null -w '%(http_code)\n' -X DELETE http://localhost/method-test
200
201
204
403
404
500
(clientside@parrot) ~)
$ curl -I http://localhost/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 15:57:50 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: keep-alive
ETag: "6a9c4efa-106d"
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes
(clientside@parrot) ~)
$ curl -I http://localhost/forbidden
HTTP/1.1 403 Forbidden
Server: nginx
Date: Wed, 09 Sep 2026 15:58:10 GMT
Content-Type: text/plain
Content-Length: 40
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07
(clientside@parrot) ~)
$ telnet 127.0.0.1 22
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^['.
SSH-2.0-OpenSSH_10.3p1 Debian-1~bpo13+1
^C
Invalid SSH identification string.
Connection closed by foreign host.
(clientside@parrot) ~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^['.
^C
telnet> quit
Connection closed.
(clientside@parrot) ~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
```

```
LISTEN 0      511          0.0.0.0:80    0.0.0.0: *    users: (('nginx', pi
d=464666, fd=5), ('nginx', pid=464665, fd=5), ('nginx', pid=464664, fd=5), ('nginx', p
id=464663, fd=5), ('nginx', pid=464662, fd=5), ('nginx', pid=464661, fd=5), ('nginx', p
id=464659, fd=5), ('nginx', pid=464658, fd=5), ('nginx', pid=464657, fd=5), ('nginx',
pid=464656, fd=5), ('nginx', pid=464655, fd=5), ('nginx', pid=464654, fd=5), ('nginx
', pid=464653, fd=5))
LISTEN 0      511          (::):80       (::): *    users: (('nginx', pi
d=464666, fd=6), ('nginx', pid=464665, fd=6), ('nginx', pid=464664, fd=6), ('nginx', p
id=464663, fd=6), ('nginx', pid=464662, fd=6), ('nginx', pid=464661, fd=6), ('nginx',
pid=464659, fd=6), ('nginx', pid=464658, fd=6), ('nginx', pid=464657, fd=6), ('nginx',
pid=464656, fd=6), ('nginx', pid=464655, fd=6), ('nginx', pid=464654, fd=6), ('nginx
', pid=464653, fd=6))
(Leo_01@parrot) ~
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
GET / HTTP/1.1
Host: localhost
Connection: close

HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 16:01:57 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: close
ETag: '6a9c4efa-106d'
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>

  <!-- Makes the website responsive on phones, tablets and computers -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>Web Infrastructure Assessment</title>

  <!-- Link the external CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
  <!-- Main header -->
  <header class='hero'>
    <div class='container'>

      <p class='server-status'>• NGINX SERVER ONLINE</p>

      <h1>Web Infrastructure Assessment</h1>

      <p class='subtitle'>
        Building and Securing a Mini Production-Ready Web Infrastruct
ure
      </p>
    </div>
  </header>
</body>
</html>
```

```
(leo_01@parrot) (~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
GET /missing-page HTTP/1.1
Host: localhost
Connection: close

HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 16:02:52 GMT
Content-Type: text/html
Content-Length: 662
Connection: close
ETag: '6a9c5478-296'
X-Module: Web-Infrastructure
X-Group: Group-07

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>404 - Page Not Found</title>

  <!-- Connect to the existing CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
  <main class='error-page'>
    <div class='error-content'>
      <h1>404</h1>

      <h2>Page Not Found</h2>

      <p>
        Sorry, the page you are looking for does not exist.
      </p>

      <a href='/' class='home-button'>
        Back to Home
      </a>

    </div>

  </main>

</body>
</html>
Connection closed by foreign host.
(lleo_01@parrot) (~)
$ https://jsonplaceholder.typicode.com
bash: https://jsonplaceholder.typicode.com: No such file or directory
(x) (leo_01@parrot) (~)
$ curl -i https://jsonplaceholder.typicode.com/posts/1
```

12. Telnet Port Testing and Raw HTTP

Telnet was used to test whether services were listening on their expected ports. Port 22 was tested for the SSH service and port 80 for the NGINX HTTP service. When a service was running and listening on its port, the connection was accepted. When NGINX was stopped, the connection to port 80 was refused. This comparison demonstrates that an open port depends on a running service listening for connections. A raw HTTP request was also entered manually through the Telnet connection. The response can be divided into four important parts. The **status line** contains the HTTP version and status code. The **response headers** provide information about the server and returned content. A **blank**

line separates the headers from the content, and the **response body** contains the actual HTML returned by the web server. This demonstrates the structure of an HTTP response without relying on a web browser.

```
(clientside@parrot) ~)
$ curl -s -o /dev/null -w '%{http_code}\n' http://localhost/
curl -s -o /dev/null -w '%{http_code}\n' -X POST http://localhost/method-test
curl -s -o /dev/null -w '%{http_code}\n' -X DELETE http://localhost/method-test
curl -s -o /dev/null -w '%{http_code}\n' http://localhost/forbidden
curl -s -o /dev/null -w '%{http_code}\n' http://localhost/does-not-exist
curl -s -o /dev/null -w '%{http_code}\n' http://localhost/error500
200
201
204
403
404
500
(clientside@parrot) ~)
$ curl -I http://localhost/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 15:57:50 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: keep-alive
ETag: "6a9c4efa-106d"
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes
(clientside@parrot) ~)
$ curl -I http://localhost/forbidden
HTTP/1.1 403 Forbidden
Server: nginx
Date: Wed, 09 Sep 2026 15:58:10 GMT
Content-Type: text/plain
Content-Length: 40
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07
(clientside@parrot) ~)
$ telnet 127.0.0.1 22
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^['.
SSH-2.0-OpenSSH_10.3p1 Debian-1~bpo13+1
^C(quit
Invalid SSH identification string.
Connection closed by foreign host.
(clientside@parrot) ~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^['.
^C)
telnet> quit
Connection closed.
(clientside@parrot) ~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
```

```

(leo_01@parrot) (~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
GET /missing-page HTTP/1.1
Host: localhost
Connection: close

HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 16:02:52 GMT
Content-Type: text/html
Content-Length: 662
Connection: close
ETag: '6a9c5478-296'
X-Module: Web-Infrastructure
X-Group: Group-07

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>404 - Page Not Found</title>

  <!-- Connect to the existing CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
  <main class='error-page'>
    <div class='error-content'>
      <h1>404</h1>

      <h2>Page Not Found</h2>

      <p>
        Sorry, the page you are looking for does not exist.
      </p>

      <a href='/' class='home-button'>
        Back to Home
      </a>

    </div>

  </main>

</body>
</html>
Connection closed by foreign host.
(leo_01@parrot) (~)
$ https://jsonplaceholder.typicode.com
bash: https://jsonplaceholder.typicode.com: No such file or directory
(x) (leo_01@parrot) (~)
$ curl -i https://jsonplaceholder.typicode.com/posts/1

```

13. Public API Interaction

The public JSONPlaceholder API was tested with different HTTP requests. The evidence includes GET requests for post data and a POST request containing JSON data. The responses include JSON fields such as userId, id, title and body, together with HTTP response headers. Three different API requests were tested. For each request, the HTTP method, endpoint, response status and returned JSON

were examined. GET requests retrieved existing data from the API, while the POST request submitted JSON data and produced a response representing the newly created resource. The returned JSON contained fields such as `userId`, `id`, `title` and `body`, demonstrating how structured data is exchanged through a REST API.

```
(x) (leo_01@parrot) (~)
$ curl -i https://jsonplaceholder.typicode.com/posts/1
HTTP/2 200
date: Wed, 09 Sep 2026 16:03:44 GMT
content-type: application/json; charset=utf-8
content-length: 292
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"124-yI KdLzqO5gfBrJFr cdJ8YqOL GnU"
expires: -1
nel: {"report_to": "heroku-nel", "response_headers": ("Via"), "max_age": 3600, "success_fraction": 0.01, "failure_fraction": 0.1}
pragma: no-cache
report-to: {"group": "heroku-nel", "endpoints": [{"url": "https://nel.heroku.com/reports?s=LAAhJwDEUDQpqnSp%2B9zUDTn32Q%2F1E6%2BPLSiFI DH2SeY%3D\u0026;id=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1785194665"}], "max_age": 3600}
reporting-endpoints: heroku-nel="https://nel.heroku.com/reports?s=LAAhJwDEUDQpqnSp%2B9zUDTn32Q%2F1E6%2BPLSiFI DH2SeY%3D\u0026;id=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1785194665"
server: cloudflare
vary: Origin, Accept-Encoding
via: 2.0 heroku-router
x-content-type-options: nosniff
x-powered-by: Express
x-ratelimit-limit: 1000
x-ratelimit-remaining: 999
x-ratelimit-reset: 1785194723
age: 8052
accept-ranges: bytes
cf-cache-status: HIT
cf-ray: a3876cb91b832857-MRU
alt-svc: h3=":443"; ma=86400

{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
}
(leo_01@parrot) (~)
$ curl -s https://jsonplaceholder.typicode.com/posts/1
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
}
(leo_01@parrot) (~)
$ curl -i "https://jsonplaceholder.typicode.com/posts?userId=1"
HTTP/2 200
date: Wed, 09 Sep 2026 16:04:28 GMT
content-type: application/json; charset=utf-8
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"aa6-j 2NSH739I 9uq40Oy wF Mn7Y0C/ I Y"
expires: -1
nel: {"report_to": "heroku-nel", "response_headers": ("Via"), "max_age": 3600, "success_fraction": 0.01, "failure_fraction": 0.1}
```

```
(leo_01@parrot) (~)
$ curl -i 'https://jsonplaceholder.typicode.com/posts?userId=1'
HTTP/2 200
date: Wed, 09 Sep 2026 16:04:28 GMT
content-type: application/json; charset=utf-8
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"aa6-j2NSH739l9uq40OywFMh7Y0C/IY"
expires: -1
nel: ({'report_to': 'heroku-nel', 'response_headers': ('Via'), 'max_age': 3600, 'success_fraction': 0.01, 'failure_fraction': 0.1})
pragma: no-cache
report-to: ({'group': 'heroku-nel', 'endpoints': ({'url': 'https://nel.heroku.com/reports?s=RRzBUBF9XeoZxcy%2BvSZ%2ButWgA4CEum%2FWfqjIDeWlIzE%3D\u0026sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d\u0026ts=1788623023'})}, {'max_age': 3600})
reporting-endpoints: heroku-nel='https://nel.heroku.com/reports?s=RRzBUBF9XeoZxcy%2BvSZ%2ButWgA4CEum%2FWfqjIDeWlIzE%3D&sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1788623023'
server: cloudflare
vary: Origin, Accept-Encoding
via: 2.0 heroku-router
x-content-type-options: nosniff
x-powered-by: Express
x-ratelimit-limit: 1000
x-ratelimit-remaining: 999
x-ratelimit-reset: 1788623032
cf-cache-status: REVALIDATED
cf-ray: a3876dc3fe045ec4-MRU
alt-svc: h3=':443'; ma=86400

(
  (
    'userId': 1,
    'id': 1,
    'title': 'sunt aut facere repellat provident occaecati excepturi optio reprehenderit',
    'body': 'quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto'
  ),
  (
    'userId': 1,
    'id': 2,
    'title': 'qui est esse',
    'body': 'est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non debitis possimus qui neque nisi nulla'
  ),
  (
    'userId': 1,
    'id': 3,
    'title': 'ea molestias quasi exercitationem repellat qui ipsa sit aut',
    'body': 'et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut'
  ),
  (
    'userId': 1,
    'id': 4,
    'title': 'eum et est occaecati',
    'body': 'ullam et saepe reiciendis voluptatem adipisci\nsit amet autem as
```



```
(leo_01@parrot) (~)
└─$ curl -i -X POST https://jsonplaceholder.typicode.com/posts \
  -H 'Content-Type: application/json' \
  -d '{"title": "Web Infrastructure", "body": "API test", "userId": 1}'
HTTP/2 201
date: Wed, 09 Sep 2026 16:04:50 GMT
content-type: application/json; charset=utf-8
content-length: 85
location: https://jsonplaceholder.typicode.com/posts/101
access-control-allow-credentials: true
access-control-expose-headers: Location
cache-control: no-cache
etag: W/"55-hGNI EySoh24h5vZJhSkZBZXt9FY"
expires: -1
nel: {"report_to": "heroku-nel", "response_headers": ("VIA"), "max_age": 3600, "success_fraction": 0.01, "failure_fraction": 0.1}
pragma: no-cache
report-to: {"group": "heroku-nel", "endpoints": [{"url": "https://nel.heroku.com/reports?s=tVbYslXlYrBWAN6lmYuiTAEa7QDxakUI0%2Fj7j%2BCTLD0%3D\u0026sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d\u0026ts=1788969890"}], "max_age": 3600}
reporting-endpoints: heroku-nel="https://nel.heroku.com/reports?s=tVbYslXlYrBWAN6lmYuiTAEa7QDxakUI0%2Fj7j%2BCTLD0%3D&sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1788969890"
server: cloudflare
vary: Origin, X-HTTP-Method-Override, Accept-Encoding
via: 2.0 heroku-router
x-content-type-options: nosniff
x-powered-by: Express
x-ratelimit-limit: 1000
x-ratelimit-remaining: 998
x-ratelimit-reset: 1788969907
cf-cache-status: DYNAMIC
cf-ray: a3876e53fa143111-MRU
alt-svc: h3=":443"; ma=86400

{
  "title": "Web Infrastructure",
  "body": "API test",
  "userId": 1,
  "id": 101
}
(leo_01@parrot) (~)
└─$ curl -i https://jsonplaceholder.typicode.com/post/1
HTTP/2 404
date: Wed, 09 Sep 2026 16:05:15 GMT
content-type: application/json; charset=utf-8
content-length: 2
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"2-vyGp6PvFo4RvsFtPolWeCReylC8"
expires: -1
nel: {"report_to": "heroku-nel", "response_headers": ("VIA"), "max_age": 3600, "success_fraction": 0.01, "failure_fraction": 0.1}
pragma: no-cache
report-to: {"group": "heroku-nel", "endpoints": [{"url": "https://nel.heroku.com/reports?s=MPfNHczX9U4AsfuUijzxLF2S3dgyNQ9lgFc7ZSlM48%3D\u0026sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d\u0026ts=1788969915"}], "max_age": 3600}
reporting-endpoints: heroku-nel="https://nel.heroku.com/reports?s=MPfNHczX9U4AsfuUijzxLF2S3dgyNQ9lgFc7ZSlM48%3D&sid=e11707d5-02a7-43ef-b45e-2cf4d2036f7d&ts=1788969915"
server: cloudflare
vary: Origin, Accept-Encoding
```



```

$ curl -i https://jsonplaceholder.typicode.com/posts/1
HTTP/2 200
date: Wed, 09 Sep 2026 16:05:24 GMT
content-type: application/json; charset=utf-8
content-length: 292
access-control-allow-credentials: true
cache-control: max-age=43200
etag: W/"124-yIKdLzqO5gfBrJFr cdJ8Yq0LGnU"
expires: -1
nel: ({'report_to': 'heroku-nel', 'response_headers': ('Via'), 'max_age': 3600, 'success_fraction': 0.01, 'failure_fraction': 0.1})
pragma: no-cache
report-to: ({'group': 'heroku-nel', 'endpoints': ({'url': 'https://nel.heroku.com/reports?s=LAAhjwDEUDQPqnSp%2B9zUDTn32Q%2F1E6%2BPLSiFI DH2SeY%3D\u0026ts=1785194665'})}, {'max_age': 3600})
reporting-endpoints: heroku-nel='https://nel.heroku.com/reports?s=LAAhjwDEUDQPqnSp%2B9zUDTn32Q%2F1E6%2BPLSiFI DH2SeY%3D\u0026ts=1785194665'
server: cloudflare
vary: Origin, Accept-Encoding
via: 2.0 heroku-router
x-content-type-options: nosniff
x-powered-by: Express
x-ratelimit-limit: 1000
x-ratelimit-remaining: 999
x-ratelimit-reset: 1785194723
age: 8152
accept-ranges: bytes
cf-cache-status: HIT
cf-ray: a3876f2928b63111-MRU
alt-svc: h3=':443'; ma=86400

{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
}
$ sudo nano /etc/nginx/sites-available/webinfrastructure
$ sudo nano /etc/nginx/sites-available/webinfrastructure
$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
$ sudo systemctl reload nginx
$ curl -i http://localhost/
HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 16:10:38 GMT
Content-Type: text/html
Content-Length: 146
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

<html>
<head><title>404 Not Found</title></head>

```

ERROR HANDLING

An incorrect API request was deliberately made to demonstrate error handling. The returned response was examined to identify the problem. The cause of the error was then corrected by fixing the request, after which the API request was sent again. The successful response from the corrected request confirmed that the problem had been resolved. This process demonstrates the importance of using HTTP status information and response data when diagnosing API errors.

14. Troubleshooting a Deliberate Infrastructure Fault

Stage 1 — Identify the problem: A deliberate web infrastructure failure was created by stopping the NGINX service. The website could no longer be reached normally.

Stage 2 — Investigate: The NGINX service status and port 80 were checked. The service was shown as inactive and the connection to port 80 was refused.

Stage 3 — Identify the cause: The evidence showed that NGINX was not running. Because the web server was stopped, it could not listen for HTTP connections on port 80.

Stage 4 — Apply the fix: NGINX was started again to restore the HTTP service.

Stage 5 — Verify the solution: The service status was checked again and showed that NGINX was active. The HTTP service was then tested again to confirm that the web infrastructure was working normally.

```
(leo_01@parrot) (~)
└─$ curl -I http://localhost/
HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 16:10:38 GMT
Content-Type: text/html
Content-Length: 146
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx</center>
</body>
</html>

(lleo_01@parrot) (~)
└─$ curl -I http://localhost/
HTTP/1.1 404 Not Found
Server: nginx
Date: Wed, 09 Sep 2026 16:11:24 GMT
Content-Type: text/html
Content-Length: 146
Connection: keep-alive
X-Module: Web-Infrastructure
X-Group: Group-07

<html>
<head><title>404 Not Found</title></head>
<body>
<center><h1>404 Not Found</h1></center>
<hr><center>nginx</center>
</body>
</html>

(lleo_01@parrot) (~)
└─$ sudo tail -n 20 /var/log/nginx/error.log
2026/09/09 20:10:38 (error) 464949#464949: *3 open() "/var/www/wrong-root/404.html" failed (2: No such file or directory), client: ::1, server: localhost, request: "GET / HTTP/1.1", host: "localhost"
2026/09/09 20:11:24 (error) 464951#464951: *4 open() "/var/www/wrong-root/404.html" failed (2: No such file or directory), client: ::1, server: localhost, request: "GET / HTTP/1.1", host: "localhost"

(lleo_01@parrot) (~)
└─$ systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Wed 2026-09-09 20:01:13 +04; 12min ago
     Invocation: ce3de289867e43b09d67840bef509833
       Docs: man:nginx(8)
    Process: 464650 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 464651 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
    Process: 464947 ExecReload=/usr/sbin/nginx -g daemon on; master_process on; -s reload (code=exited, status=0/SUCCESS)
   Main PID: 464653 (nginx)
      Tasks: 13 (limit: 18537)
     Memory: 11M (peak: 23.9M)
```

```
(leo_01@parrot) (~)
└─$ sudo ss -tlnp | grep :80
LISTEN 0      511          0.0.0.0:80          0.0.0.0:*        users:(('nginx', pi
d=464961, fd=5), ('nginx', pid=464960, fd=5), ('nginx', pid=464958, fd=5), ('nginx', p
id=464957, fd=5), ('nginx', pid=464956, fd=5), ('nginx', pid=464955, fd=5), ('nginx',
pid=464954, fd=5), ('nginx', pid=464953, fd=5), ('nginx', pid=464952, fd=5), ('nginx'
, pid=464951, fd=5), ('nginx', pid=464950, fd=5), ('nginx', pid=464949, fd=5), ('ngi
nx', pid=464653, fd=5))
LISTEN 0      511          (::):80            (::):*        users:(('nginx', pi
d=464961, fd=6), ('nginx', pid=464960, fd=6), ('nginx', pid=464958, fd=6), ('nginx', p
id=464957, fd=6), ('nginx', pid=464956, fd=6), ('nginx', pid=464955, fd=6), ('nginx'
, pid=464954, fd=6), ('nginx', pid=464953, fd=6), ('nginx', pid=464952, fd=6), ('ngi
nx', pid=464951, fd=6), ('nginx', pid=464950, fd=6), ('nginx', pid=464949, fd=6), ('ngi
nx', pid=464653, fd=6))
(lleo_01@parrot) (~)
└─$ sudo nginx -T | grep -n 'root'
197:         root /var/www/wrong-root;
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(lleo_01@parrot) (~)
└─$ ls -ld /var/www/webinfrastructure
ls -ld /var/www/wrong-root
drwxr-xr-x 1 www-data www-data 50 Sep  9 18:39 /var/www/webinfrastructure
ls: cannot access '/var/www/wrong-root': No such file or directory
(x)(leo_01@parrot) (~)
└─$ sudo nano /etc/nginx/sites-available/webinfrastructure
(lleo_01@parrot) (~)
└─$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(lleo_01@parrot) (~)
└─$ sudo systemctl reload nginx
(lleo_01@parrot) (~)
└─$ curl -i http://localhost/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 16:22:18 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: keep-alive
ETag: "6a9c4efa-106d"
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes

<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>

  <!-- Makes the website responsive on phones, tablets and computers -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>Web Infrastructure Assessment</title>

  <!-- Link the external CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>
```

```

        <span>04</span>
        <h3>SSH Port</h3>
        <p>Port 22</p>
    </div>
</div>
</section>
<!-- Technologies -->
<section class='card'>
    <h2>Technologies Demonstrated</h2>
    <div class='tags'>
        <span>Linux</span>
        <span>SSH</span>
        <span>NGINX</span>
        <span>HTTP</span>
        <span>Telnet</span>
        <span>curl</span>
        <span>REST API</span>
    </div>
</section>
</main>
<!-- Footer -->
<footer>
    <div class='container'>
        <p>Web Infrastructure Assessment &copy; 2026</p>
        <p>Hosted by NGINX on Parrot OS</p>
    </div>
</footer>
</body>
</html>
(Leo_01@parrot) ~$ curl -I http://localhost/
HTTP/1.1 200 OK
Server: nginx
Date: Wed, 09 Sep 2026 16:22:31 GMT
Content-Type: text/html
Content-Length: 4205
Last-Modified: Sat, 05 Sep 2026 17:18:50 GMT
Connection: keep-alive
ETag: '6a9c4efa-106d'
X-Module: Web-Infrastructure
X-Group: Group-07
Accept-Ranges: bytes
(Leo_01@parrot) ~$
```

15. Summary

The evidence demonstrates Linux account setup, password and key-based SSH authentication, SCP and SFTP transfers, NGINX configuration, HTTP method and status testing, Telnet port testing, raw HTTP inspection, public API interaction and troubleshooting of an NGINX service failure.

```

[clientside@parrot] (~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^A'.
^A)
telnet> quit
Connection closed.
[clientside@parrot] (~)
$ sudo systemctl stop nginx
(sudo) password for clientside:
clientside is not in the sudoers file.
[clientside@parrot] (~)
$ exit
logout
[leo_01@parrot] (~)
$ sudo systemctl stop nginx
Warning: The unit file, source configuration file or drop-ins of nginx.service
changed on disk. Run 'systemctl daemon-reload' to reload units.
[leo_01@parrot] (~)
$ systemctl daemon-reload
[leo_01@parrot] (~)
$ sudo systemctl stop nginx
[leo_01@parrot] (~)
$ systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset:
disabled)
   Active: inactive (dead) since Wed 2026-09-09 20:00:05 +04; 39s ago
     Duration: 3d 21h 44min 32.861s
  Invocation: 38b50667c8474bfdad1f7c61bb4332a6
        Docs: man:nginx(8)
    Main PID: 20047 (code=exited, status=0/SUCCESS)
   Mem peak: 24.4M
      CPU: 182ms
[leo_01@parrot] (~)
$ sudo ss -tlnp | grep :80
[leo_01@parrot] (~)
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
telnet: Unable to connect to remote host: Connection refused
[leo_01@parrot] (~)
$ sudo systemctl start nginx
[leo_01@parrot] (~)
$ systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset:
disabled)
   Active: active (running) since Wed 2026-09-09 20:01:13 +04; 21s ago
     Invocation: ce3de289867e43b09d67840bef509833
        Docs: man:nginx(8)
    Process: 464650 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_p
rocess on; (code=exited, status=0/SUCCESS)
    Process: 464651 ExecStart=/usr/sbin/nginx -g daemon on; master_process on
; (code=exited, status=0/SUCCESS)
   Main PID: 464653 (nginx)
      Tasks: 13 (limit: 18537)
     Memory: 10M (peak: 12M)
        CPU: 55ms
      CGroup: /system.slice/nginx.service

```

```
<!DOCTYPE html>
<html lang='en'>
<head>
  <meta charset='UTF-8'>

  <!-- Makes the website responsive on phones, tablets and computers -->
  <meta name='viewport' content='width=device-width, initial-scale=1.0'>

  <title>Web Infrastructure Assessment</title>

  <!-- Link the external CSS file -->
  <link rel='stylesheet' href='style.css'>
</head>
<body>

  <!-- Main header -->
  <header class='hero'>
    <div class='container'>

      <p class='server-status'>● NGINX SERVER ONLINE</p>

      <h1>Web Infrastructure Assessment</h1>

      <p class='subtitle'>Building and Securing a Mini Production-Ready Web Infrastructur
        e
      </p>

    </div>
  </header>

  <!-- Main website content -->
  <main class='container'>

    <!-- About the project -->
    <section class='card'>
      <h2>About This Project</h2>

      <p>This website has been created as part of our Web Infrastructure assessment. It is hosted locally using the NGINX web server running on Parrot OS Linux.</p>

      <p>The project demonstrates the configuration, testing and troubleshooting of essential web infrastructure services, including SSH, NGINX and HTTP.</p>
    </section>

    <!-- Group information -->
    <section class='card'>
      <h2>Our Group</h2>

      <div class='students'>
```